

Age and Gender Detection using Python & OpenCV

Aim of the Project

This project uses **Deep Learning** and **Computer Vision** to detect human faces in an image or video stream and predict the person's **age group** and **gender**. It demonstrates how AI can analyze facial features to make demographic predictions in real time, using lightweight models optimized for fast inference on laptops or edge devices.

Project Description

This project utilizes **OpenCV's Deep Neural Network (DNN)** module with pre-trained models to identify **gender (Male/Female)** and **age ranges** such as (0–2), (4–6), (25–32), (48–53), and (60–100) from a detected face region.

It combines:

- **Face Detection** using Haar Cascade or DNN models
- **Age & Gender Classification** using CNNs trained on the Caffe framework
- **Real-Time Visualization** via OpenCV window with labeled bounding boxes

This demonstrates how neural networks can extract facial features and perform high-speed inference on live video streams.

Tools & Libraries

- **Python 3.x** – core programming language
- **OpenCV** – for computer vision and image processing
- **NumPy** – for numerical computations
- **Caffe Models** – pre-trained CNNs for age and gender detection

How It Works

1. **Face Detection:** Detects face regions using Haar Cascade or DNN.
 2. **Preprocessing:** Crops and resizes faces to 227×227 pixels.
 3. **Prediction:**
 - Gender model classifies Male/Female.
 - Age model predicts approximate age range.
 4. **Display:** Draws bounding boxes and shows predicted results in real time.
-